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Q1. Scarlet Bakery bakes bagels and muffins. A dozen bagels uses 2 kg of flour and 1 hour of oven time; a dozen muffins uses 1 kg of flour and 1 hour of oven time. Each day 40 kg of flour and 30 oven hours are available, and at least 5 dozen muffins must be baked. Profit is $9 per dozen bagels and $7 per dozen muffins. Formulate an LP that maximizes daily profit.

(a) Define the decision variables. [8 pts]

x1 = dozens of bagels per day, x2 = dozens of muffins per day

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(b) Write the objective function. [8 pts]

max z = 9x1 + 7x2

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(c) Write all constraints. [14 pts]

x1 + 2x2 <= 40; x1 + x2 <= 30; x2 >= 5; x1, x2 >= 0

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Flour use is 2 kg per dozen bagels and 1 kg per dozen muffins: 2x1 + x2 <= 40.

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Q2. Convert to standard form: max z = x1 + 4x2 subject to -x1 + x2 <= 3, 3x1 + x2 >= 6, x1 >= 0, x2 free.

(a) Write the constraints as equalities with slack/surplus variables. [10 pts]

-x1 + x2 + s1 = 3; 3x1 + x2 - s2 = 6

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Slack and surplus variables must be non-negative too - add s1, s2 >= 0.

(b) Handle the free variable x2 and rewrite the objective. [10 pts]

x2 >= 0; max z = x1 + 4x2

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x2 is free: substitute x2+ - x2- instead of assuming x2 >= 0.

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Q3. Simplex: max z = 2x1 + 3x2 subject to x1 + 2x2 <= 8, 3x1 + 2x2 <= 12, x1, x2 >= 0.

(a) Choose the entering and leaving variables for the first pivot. [10 pts]

x2 enters (-3); ratios 8/2 = 4, 12/2 = 6, so s1 leaves.

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(b) Carry out pivots to optimality. [10 pts]

Pivot 2: x1 enters, x2 leaves (ratio 8).

Minimum ratio in the second pivot is 4 / 2 = 2 on the s2 row, not 4 / (1/2) = 8.

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(c) State the optimal solution and value; explain how you know it is optimal. [10 pts]

x1 = 2, x2 = 3, z = 13; optimal because all z-row coefficients are >= 0.

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Q4. Graphical method: max z = 5x1 + 5x2 subject to x1 + x2 <= 8, x1 <= 5, x2 <= 6, x1, x2 >= 0.

(a) List the corner points of the feasible region. [10 pts]

(0, 0), (5, 0), (5, 3), (2, 6), (0, 6)

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(b) Find all optimal solutions and explain. [10 pts]

The LP is degenerate, so there is no unique answer.

Multiple optima come from a parallel objective, not degeneracy (a zero basic variable).

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